

VIITASAARI, Finland

WMO: 029150

Lat: 63.0822N

Lon: 25.8586E

Elev: 433

StdP: 14.47

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-13.9	-7.6	-19.0	2.0	-13.9	-12.1	2.9	-7.5	14.3	26.2	13.0	26.9	1.1	150	0.568

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.1	79.7	64.6	76.1	63.2	73.1	61.5	67.4	75.3	65.5	72.9	63.5	70.6	5.1	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.6	92.7	70.6	62.6	86.2	69.0	60.6	80.1	67.1	32.1	75.2	30.5	72.8	29.1	70.3	75.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.3	11.0	9.7	DB	-17.1	83.5	8.7	4.2	-23.3	86.5	-28.4	88.9	-33.3	91.3	-39.6	94.3	(4)
(5)			WB	-17.1	69.6	8.5	3.2	-23.3	71.8	-28.2	73.7	-33.0	75.5	-39.2	77.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	39.4	17.6	17.6	25.7	37.3	48.4	57.7	63.3	60.1	50.6	39.1	30.4	23.6		
	DBStd	18.00	12.24	12.82	8.71	6.68	7.90	6.76	5.92	5.62	6.01	6.79	7.87	11.40		
	HDD50	5004	1005	909	752	385	126	11	0	2	66	341	587	820		
	HDD65	9431	1470	1329	1217	832	517	236	104	169	432	803	1037	1285		
	CDD50	1135	0	0	0	2	76	241	412	317	84	3	0	0		
	CDD65	88	0	0	0	0	2	16	51	19	0	0	0	0		
	CDH74	543	0	0	0	0	28	112	316	87	0	0	0	0		
	CDH80	79	0	0	0	0	1	13	56	9	0	0	0	0		
	WSAvg	4.3	4.2	4.3	4.6	4.3	4.2	4.0	3.6	3.8	4.1	4.6	4.9	4.8		
	PrecAvg	23.6	1.5	1.3	1.2	1.2	1.8	2.6	3.2	2.6	2.1	2.2	1.9	1.8		
	PrecMax	29.9	3.3	2.6	2.4	2.5	4.6	5.0	5.8	4.6	5.3	4.4	4.4	3.0		
(15) Precipitation	PrecMin	16.6	0.3	0.1	0.2	0.4	0.5	0.9	0.7	0.8	0.7	0.5	0.2	0.5		
	PrecStd	3.0	0.7	0.7	0.5	0.6	1.0	1.1	1.3	1.1	1.2	1.0	0.9	0.7		
	DB	39.0	40.8	47.7	63.6	78.2	82.2	85.1	81.0	69.2	55.5	48.7	43.0			
	MCWB	37.1	37.2	39.3	49.9	58.9	64.7	67.2	66.8	60.3	51.4	46.8	41.7			
	2% DB	36.0	37.3	43.0	56.9	71.9	77.3	81.2	76.1	65.1	52.5	45.3	39.9			
(23) Mean Coincident Wet Bulb Temperatures	MCWB	34.7	35.5	37.0	46.2	56.8	62.7	66.4	63.6	57.8	50.6	43.8	38.7			
	5% DB	34.1	34.5	39.6	52.4	66.6	73.3	77.4	73.0	62.0	50.8	42.7	37.7			
	MCWB	33.3	33.4	35.0	43.2	53.8	60.2	64.4	62.0	56.1	49.0	41.7	36.6			
	10% DB	32.5	32.8	36.6	48.6	62.1	69.9	74.1	69.9	59.4	48.8	40.3	35.7			
	MCWB	31.8	31.9	33.3	41.2	51.3	58.7	63.2	61.0	54.9	47.0	39.3	34.7			
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	37.3	37.7	40.7	52.0	61.9	67.9	71.2	68.4	61.3	53.1	47.3	42.0			
	MCDB	38.5	40.0	46.2	59.2	73.3	76.6	80.0	76.8	66.8	54.2	48.6	42.7			
	2% WB	35.0	35.6	37.8	47.8	58.4	65.1	68.4	65.6	59.1	51.1	44.0	39.1			
	MCDB	35.8	37.0	41.8	55.3	67.8	73.1	76.4	73.1	63.3	52.3	45.0	39.9			
	5% WB	33.5	33.6	35.7	44.3	55.5	62.4	66.6	63.8	57.4	49.4	41.7	36.8			
(31) Mean Daily Temperature Range	MCDB	34.1	34.3	38.6	50.8	64.8	70.7	74.1	70.3	60.9	50.5	42.7	37.5			
	10% WB	32.0	32.1	33.8	41.6	52.5	60.0	64.8	62.1	55.6	47.1	39.4	34.8			
	MCDB	32.5	32.7	36.0	47.7	60.6	67.8	72.0	68.0	58.6	48.5	40.2	35.5			
	5% DB	7.9	8.4	15.4	21.5	23.3	20.0	18.7	17.4	14.1	8.8	7.7	7.0			
	MCWBR	7.7	7.5	10.7	12.4	11.2	8.9	7.4	7.3	7.7	6.8	7.5	6.7			
(38) Clear-Sky Solar Irradiance	5% WB	7.7	8.0	12.9	19.7	20.1	17.4	16.0	14.9	12.4	8.1	7.5	6.9			
	MCWBR	7.6	7.3	9.6	12.2	10.6	8.9	7.5	7.6	7.6	6.9	7.3	6.7			
	taub	0.226	0.283	0.311	0.352	0.347	0.347	0.370	0.354	0.330	0.310	0.244	0.358			
	taud	2.012	2.131	2.204	2.334	2.454	2.479	2.429	2.465	2.520	2.398	2.102	2.641			
	Ebn at Noon	126	204	248	263	276	278	266	261	246	200	117	73			
(44) All-Sky Solar Radiation	Edh at Noon	10	21	29	32	30	30	31	28	22	16	9	6			
	RadAvg	57	233	709	1163	1528	1664	1578	1206	686	268	67	25			
	RadStd	8	50	76	134	171	131	163	132	79	50	13	3			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (39 neighbors)	+1.01	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page